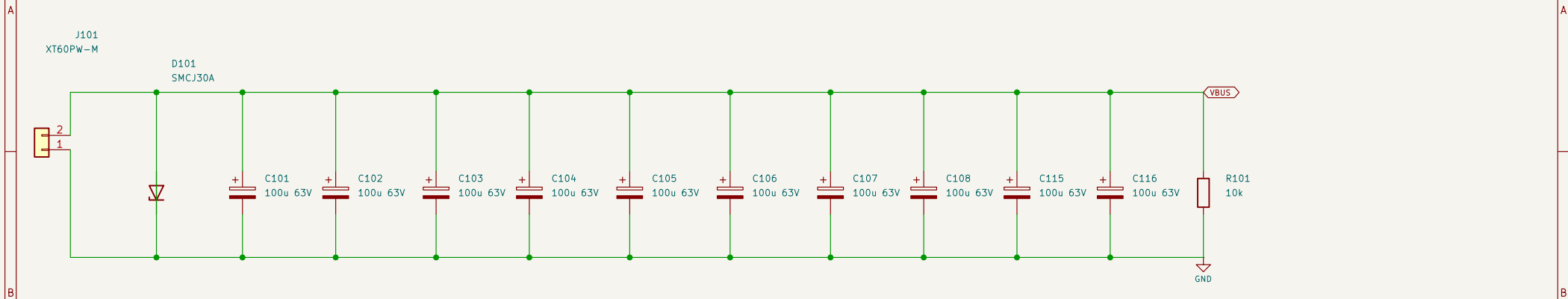


	1	2	3	4	5	6	7	8
A	BLDC–ESC / v1 / schematic draft							
B	01_dc_link XT60 / DC–link / TVS / MT60 File: schematic/01_dc_link.kicad_sch	02_bridge MOSFETs / shunts / INA241 File: schematic/02_bridge.kicad_sch	03_regen Bus clamp / dump switch File: schematic/03_regen.kicad_sch	04_power_usb Buck / power mux / 3V3 File: schematic/04_power_usb.kicad_sch				
C	05_mcu STM32G474 / reference / SWD File: schematic/05_mcu.kicad_sch	06_sensors Hall / encoder / sensor power File: schematic/06_sensors.kicad_sch	07_io_can CAN FD / UART File: schematic/07_io_can.kicad_sch	08_analog_inputs Throttle / RC PWM / digital I/O File: schematic/08_analog_inputs.kicad_sch				
D	09_safety Arm latch / PWM buffer / interlocks File: schematic/09_safety.kicad_sch	10_voltage_sensing Bus and phase ADC buffers File: schematic/10_voltage_sensing.kicad_sch	11_current_protection Hardware current and bus windows File: schematic/11_current_protection.kicad_sch	12_gate_driver DRV8353 / gate supplies / SPI File: schematic/12_gate_driver.kicad_sch				
E	13_usb_status USB–C / ESD / status LEDs File: schematic/13_usb_status.kicad_sch	14_temperature NTCs / analog rail monitor File: schematic/14_temperature.kicad_sch	15_regen_safety Chopper OCP / latch / telemetry File: schematic/15_regen_safety.kicad_sch	16_supervisor Reset / watchdog / service mode File: schematic/16_supervisor.kicad_sch				
F	X = DNP / not fitted. v1 draft; electrical review and bench validation pending.				Sheet: / File: bldc–esc.kicad_sch Title: Size: A3Date:Rev: KiCad E.D.A. 10.0.4Id: 1/17			

DC input and motor connections

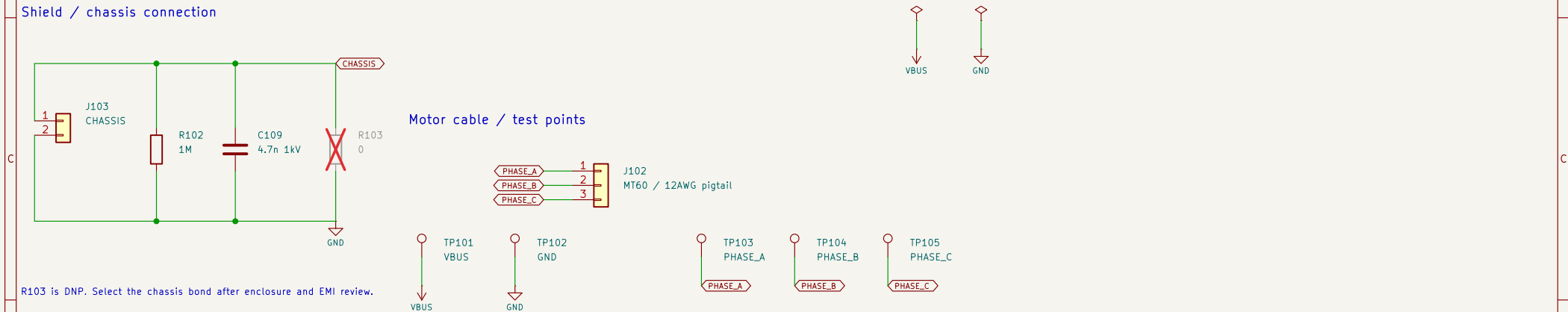
DC input and local energy storage



12–30 V operating bus. External DC-rated fuse required close to the source. No electronic reverse–polarity protection.

C101–C108, C115–C116: 10 x 100 uF / 63 V hybrid, 1,000 uF total. Bulk bank does not absorb sustained regeneration.

Shield / chassis connection



R103 is DNP. Select the chassis bond after enclosure and EMI review.

MT60 pigtail: 1=A, 2=B, 3=C. Provide strain relief.

Sheet: /01_dc_link/
File: 01_dc_link.kicad_sch

Title: DC input and motor connections

Size: A4

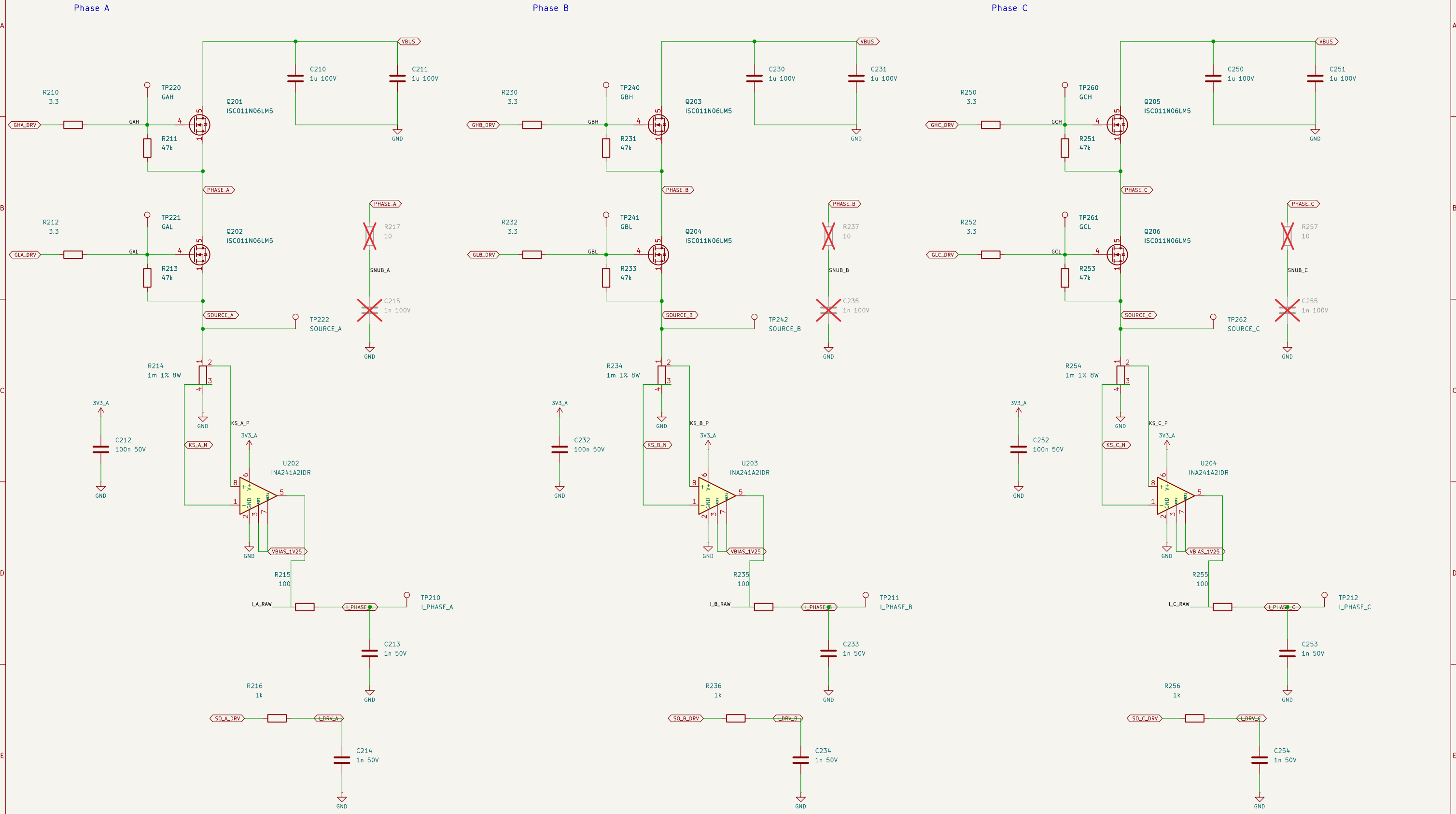
Date:

KiCad E.D.A. 10.0.4

Rev: v1–draft

Id: 2/17

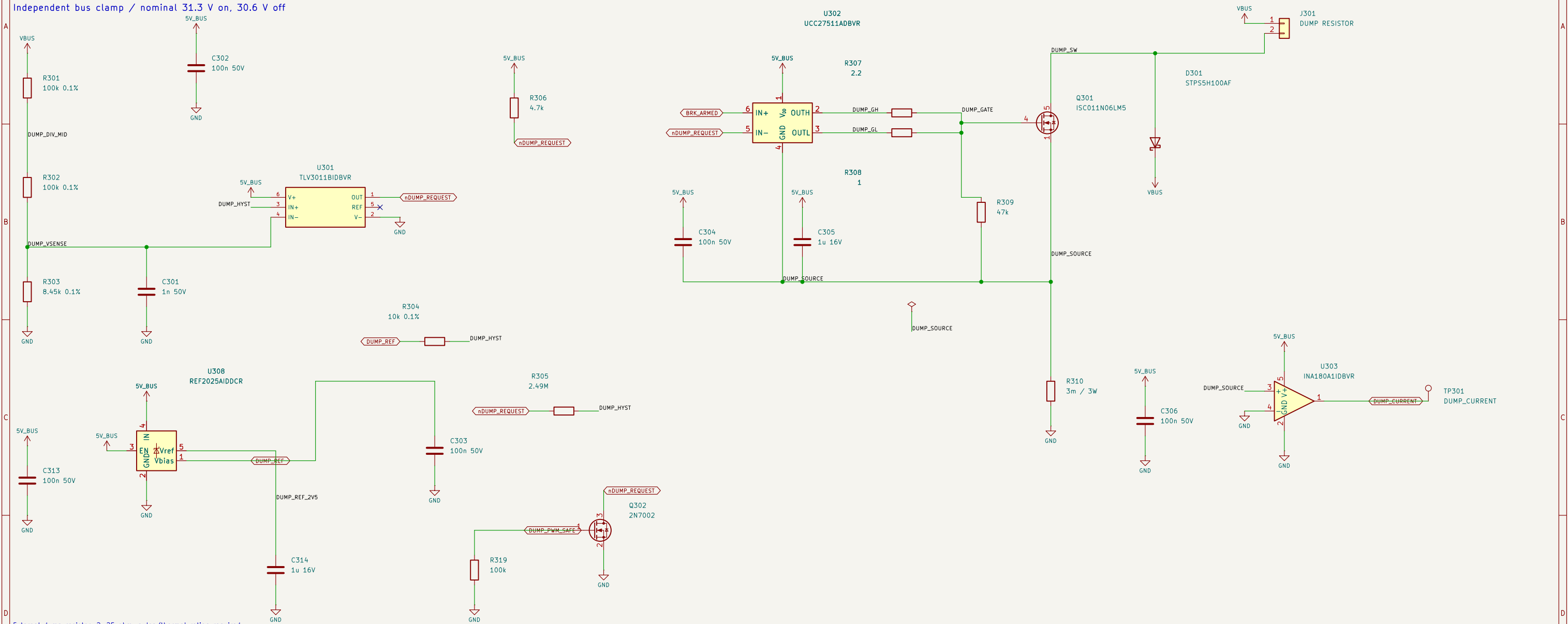
Three-phase bridge and current sensing



Main CSA: 1.25 V zero, 20 mV/A
Gate probes need local source returns.
High-side VGS: differential probe only.

Regenerative brake chopper

Independent bus clamp / nominal 31.3 V on, 30.6 V off



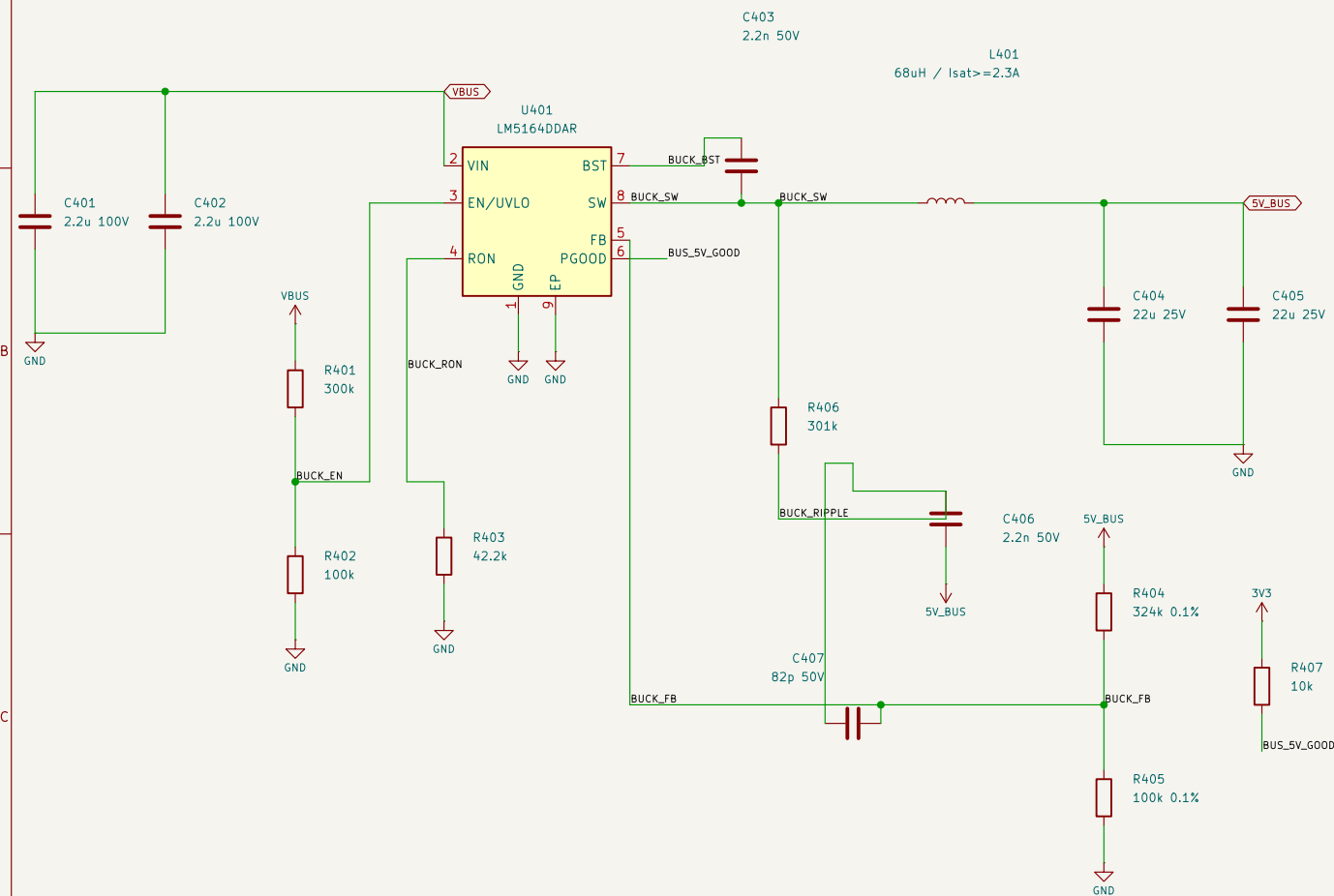
External dump resistor: 2–25 ohm; pulse/thermal rating required.
Chopper trip: about 20.8 A, power–cycle reset. Reference U308 stays bus–powered.

U302 return is Kelvin–connected to the top of R310.
Do not force a bench supply to sink regenerative current.

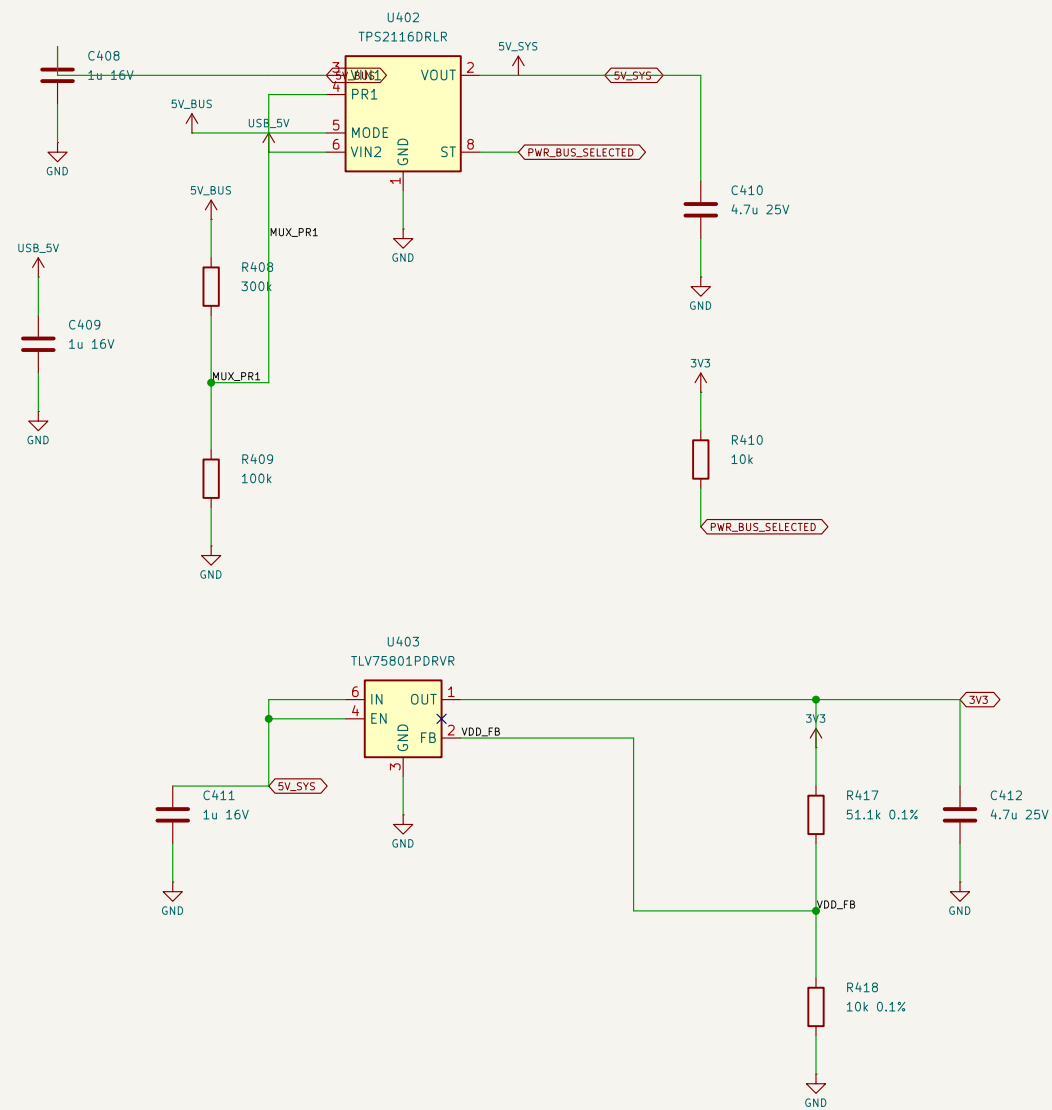
Power supplies

Bus supply / LM5164, nominal 5 V

Bus / USB selection and logic supply



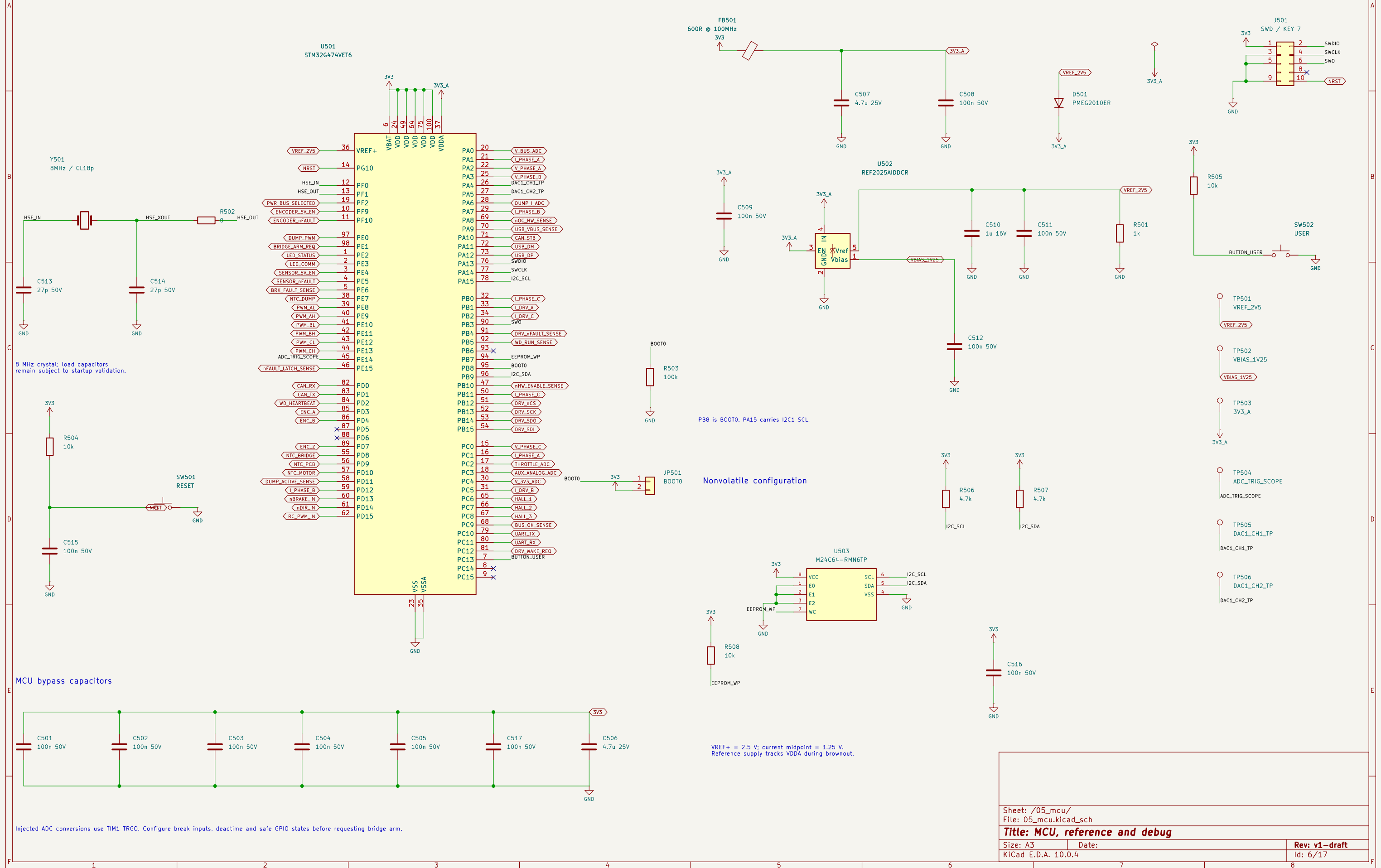
68 uH / about 300 kHz. R406, C406, C407 provide type-3 ripple injection. Verify startup and load transients.



```
3V3 net: 3.3605 V nominal (51.1k / 10k).
USB-only: bridge disarmed; sensor supplies off at boot.
```

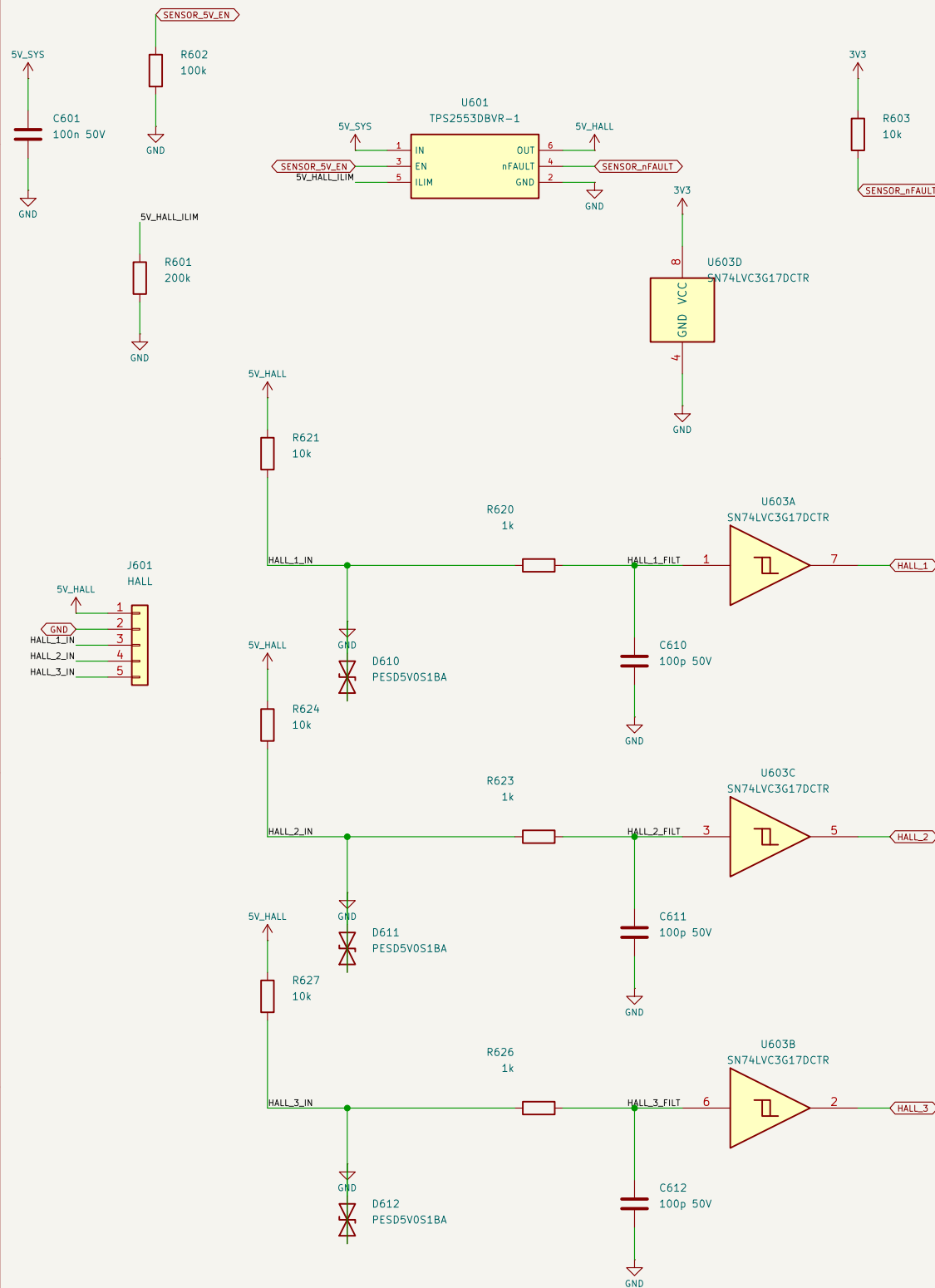
MCU, reference and debug

STM32G474VET6 / TIM1 bridge control



Hall and encoder interfaces

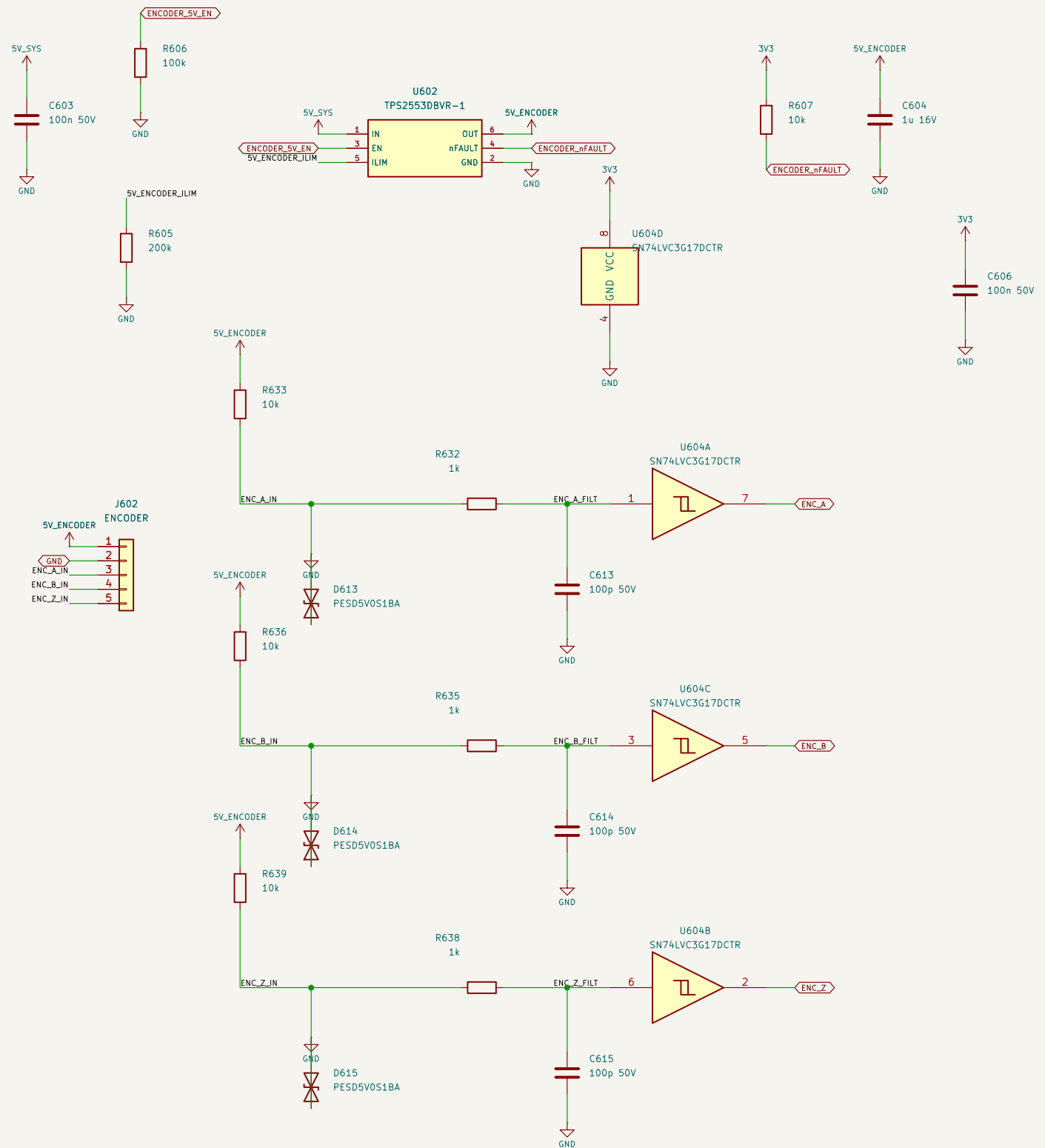
Hall interface



5 V supply: current limited, latch-off on fault.
Inputs: 3.3/5 V single-ended, not 24 V.

Connector labels define the controller pinout only. Verify the OEM motor Hall supply, ground and signal wires before connection.

Encoder interface



5 V supply: current limited, latch-off on fault.
Inputs: 3.3/5 V single-ended, not 24 V.

Sheet: /06_sensors/
File: 06_sensors.kicad_sch

Title: Hall and encoder interfaces

Size: A3

Date:

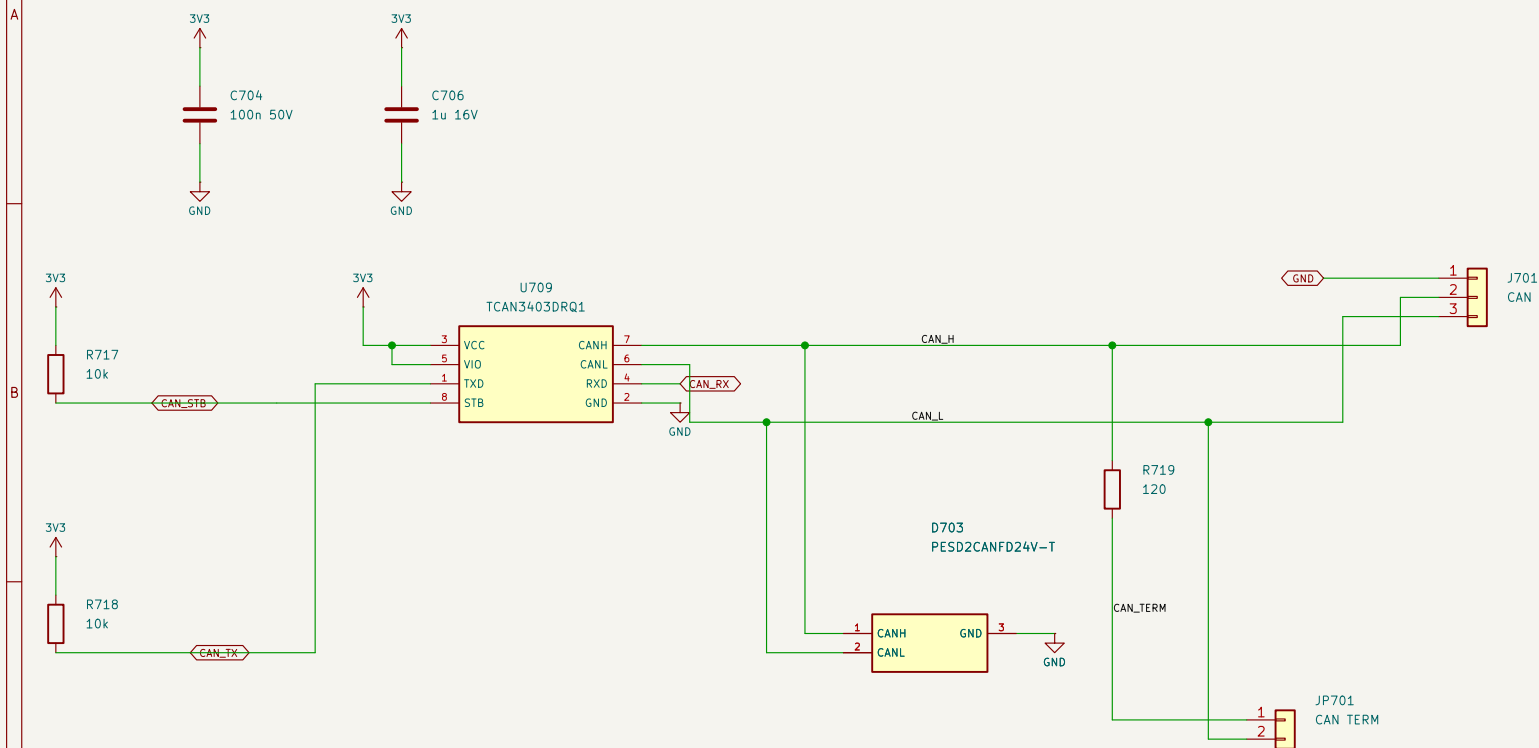
Rev: v1-draft

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Id: 7/17

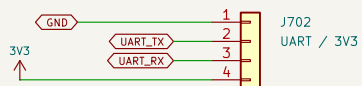
CAN FD and UART

CAN FD



JP701 fitted = 120 ohm termination. Fit termination only at the two ends of the CAN bus.

UART / 3.3 V logic



Short local 3.3 V wiring only; TX/RX are controller-side names.
Not an RS-232 or 5 V port. Supply pin is an output, not a power input.

Sheet: /07_io_can/
File: 07_io_can.kicad_sch

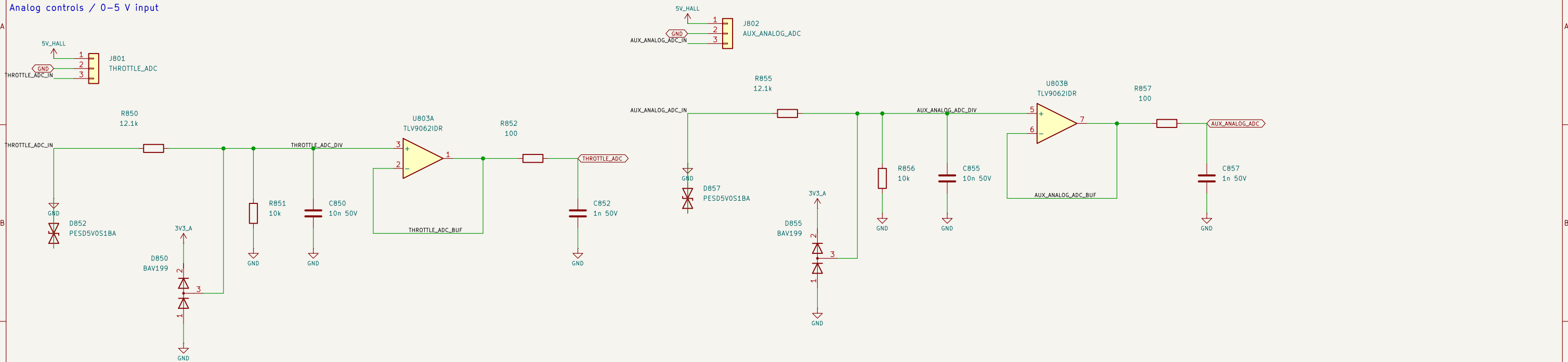
Title: CAN FD and UART

Size: A4	Date:
KiCad E.D.A. 10.0.4	

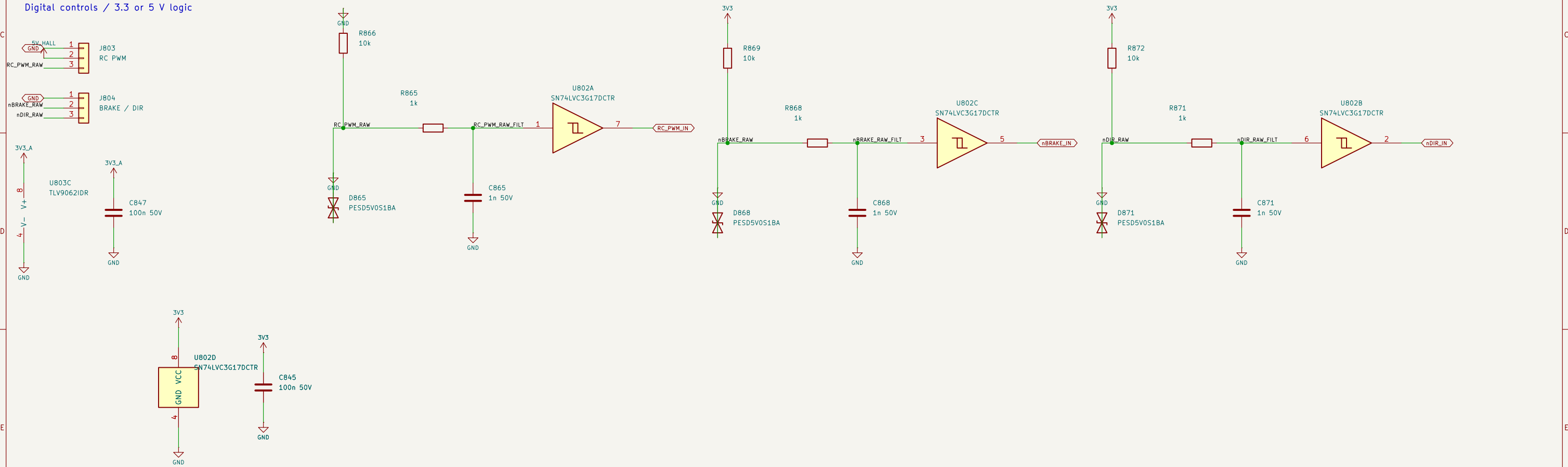
Rev: v1-draft
Id: 8/17

Analog and digital controls

Analog controls / 0–5 V input

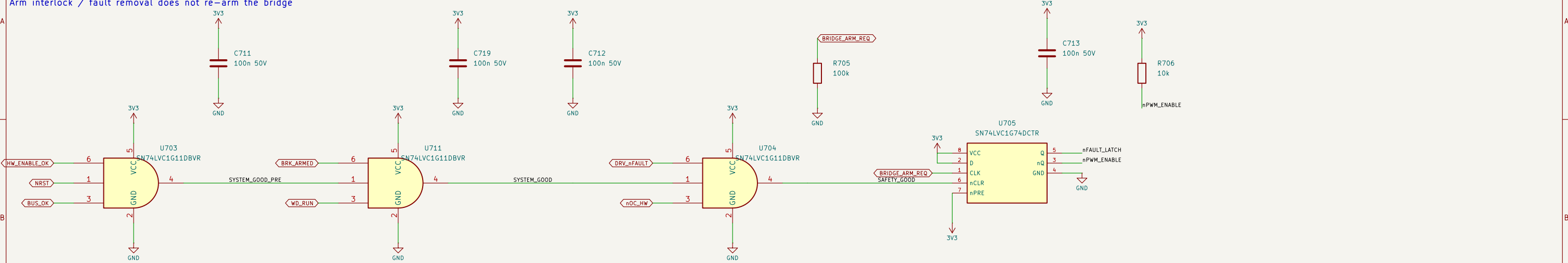


Digital controls / 3.3 or 5 V logic

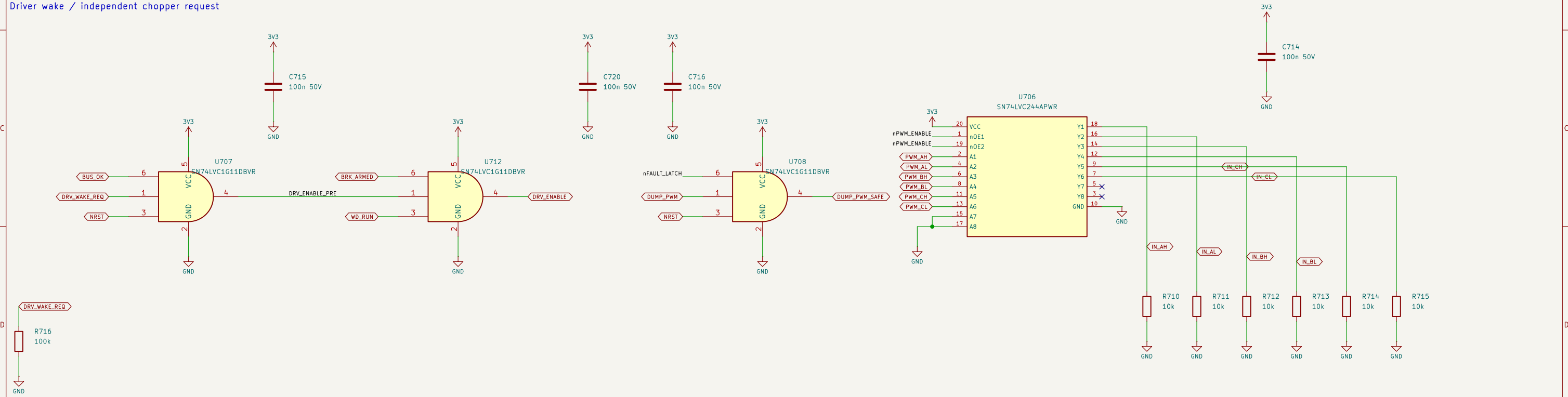


Hardware bridge interlock

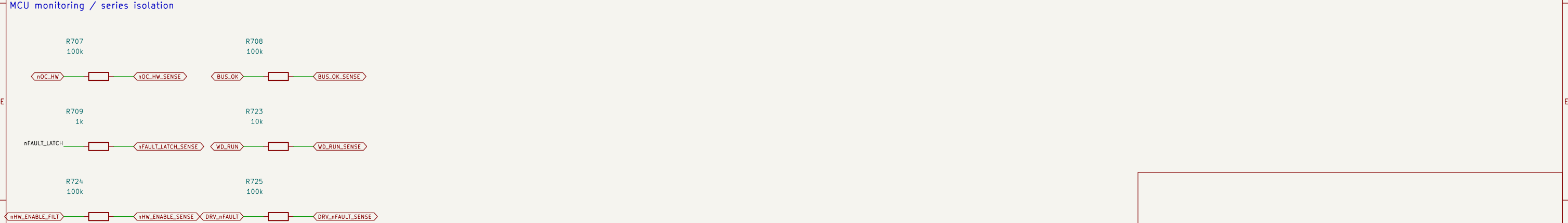
Arm interlock / fault removal does not re-arm the bridge



Driver wake / independent chopper request



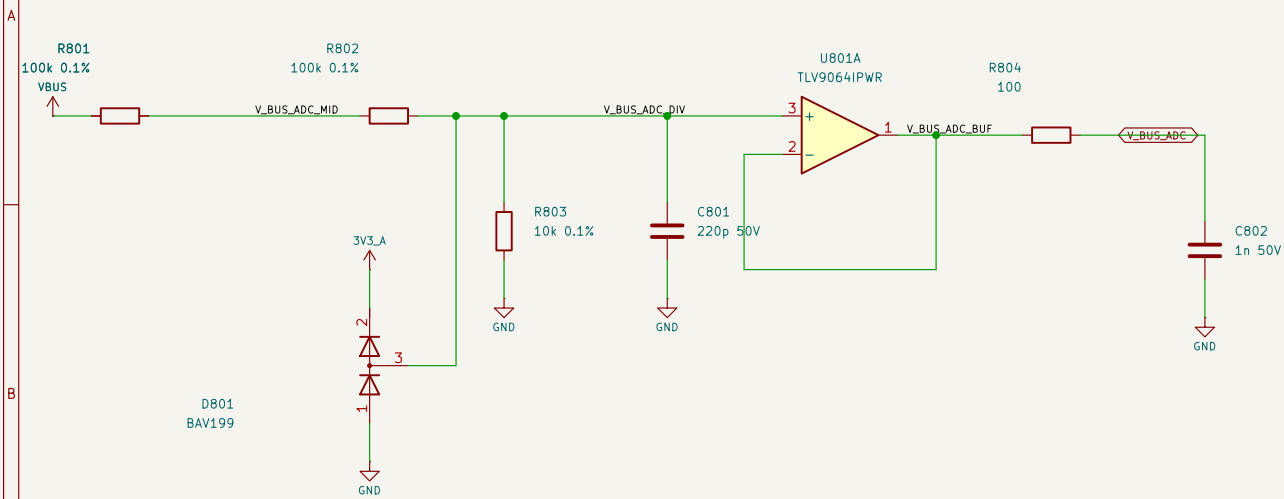
MCU monitoring / series isolation



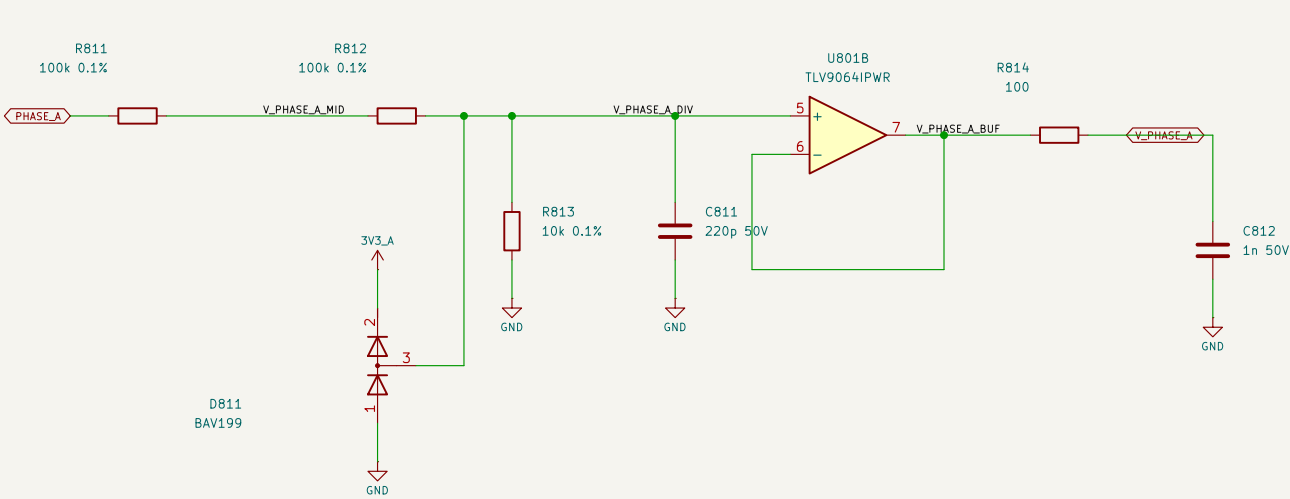
Hardware fault and enable nodes are isolated from MCU GPIO drive.
Disable UCPD dead-battery control before reading PB4.

Bus and phase voltage sensing

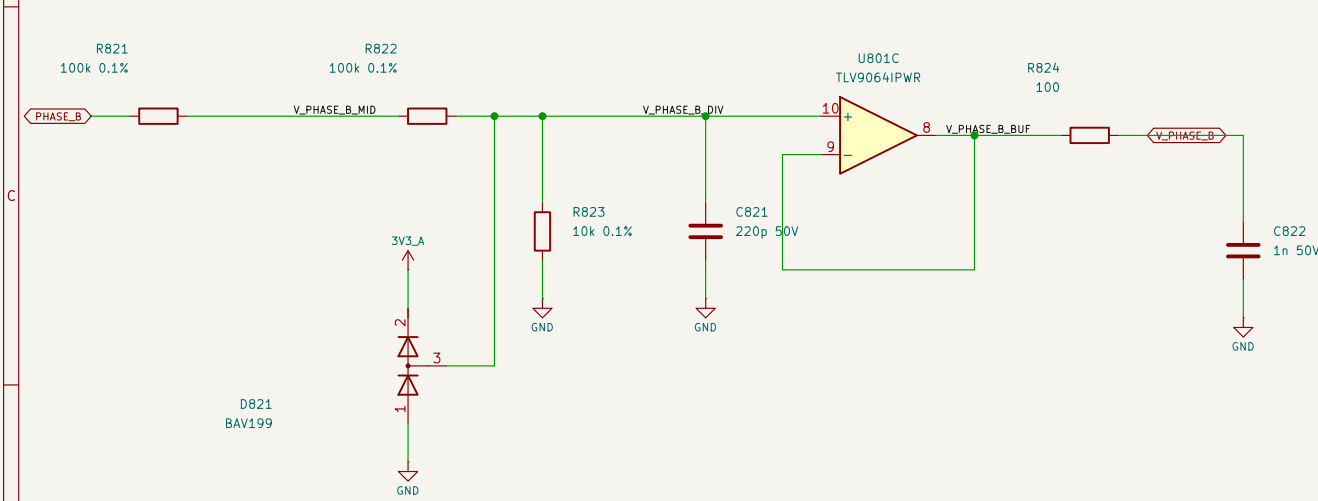
Bus voltage



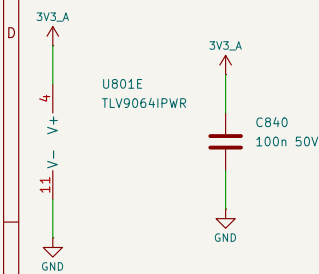
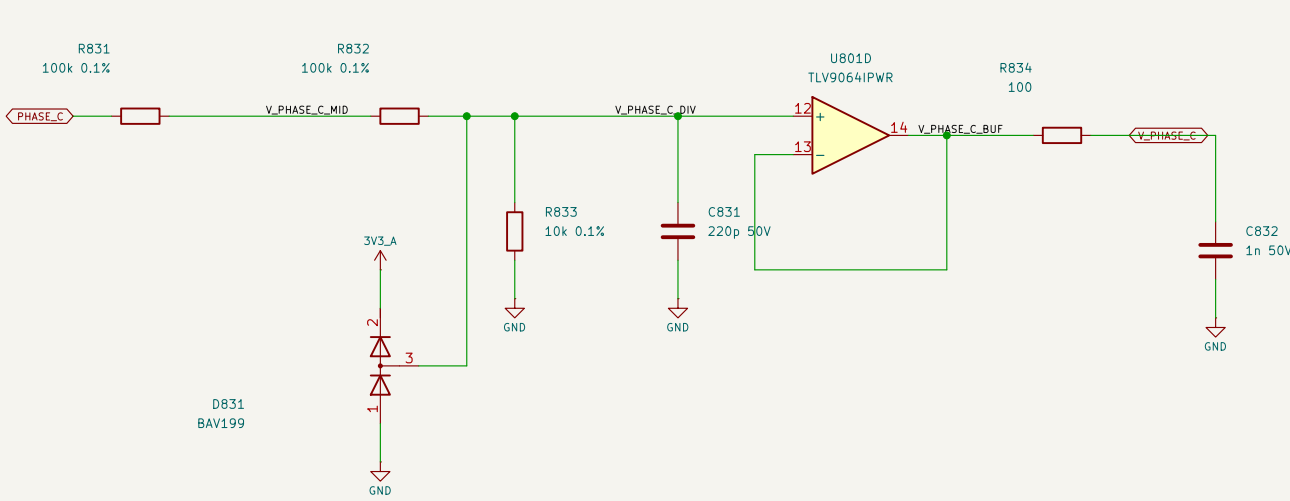
Phase A voltage



Phase B voltage



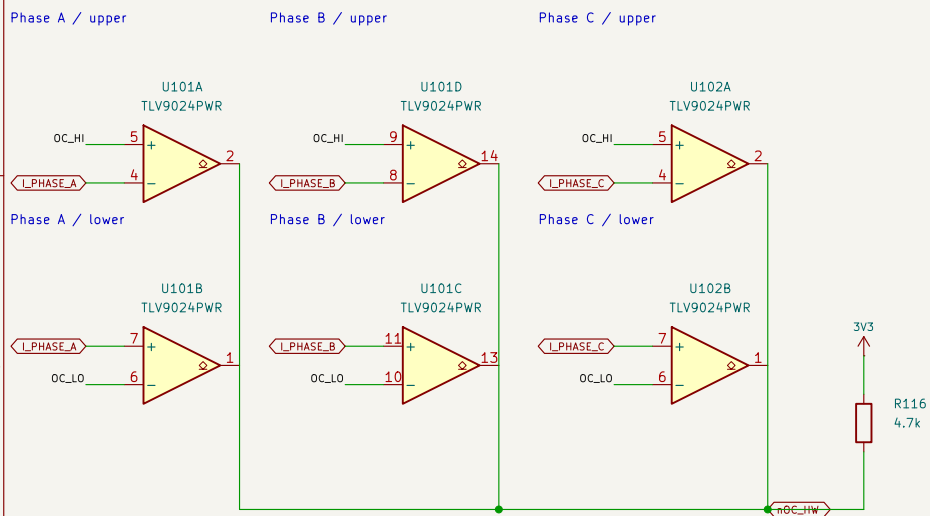
Phase C voltage



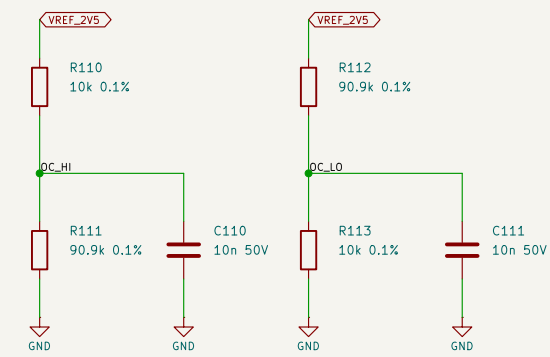
21:1 dividers; 2.5 V ADC full scale = 52.5 V at the input.
Buffers isolate the dividers from ADC sampling transients.
Normal operation remains limited to a 12–30 V bus.

Current and bus protection

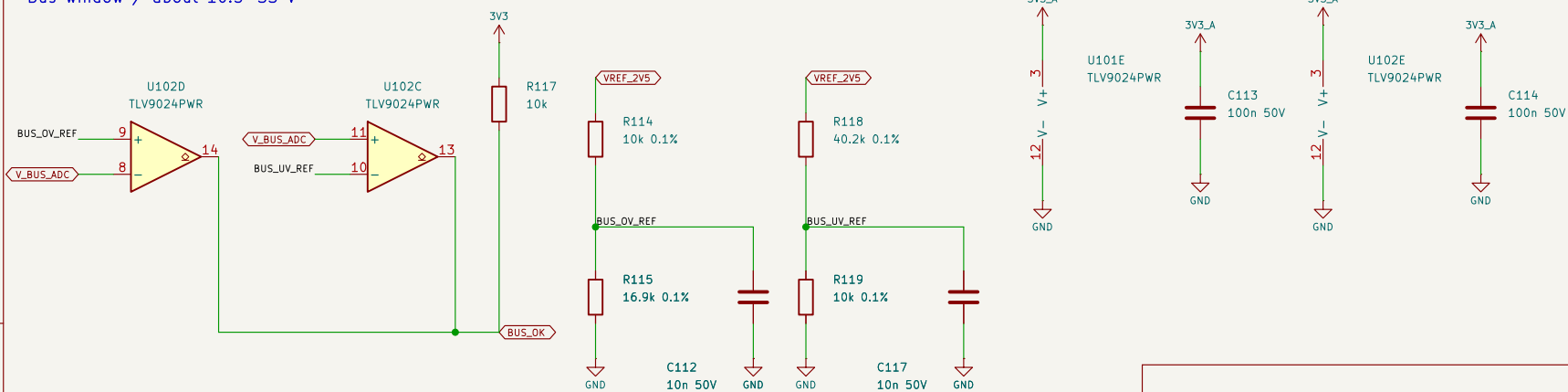
Bipolar phase-current window



Current thresholds / about ± 50 A



Bus window / about 10.5–33 V



Sheet: /11_current_protection/
File: 11_current_protection.kicad_sch

Title: Current and bus protection

Size: A4

Date:

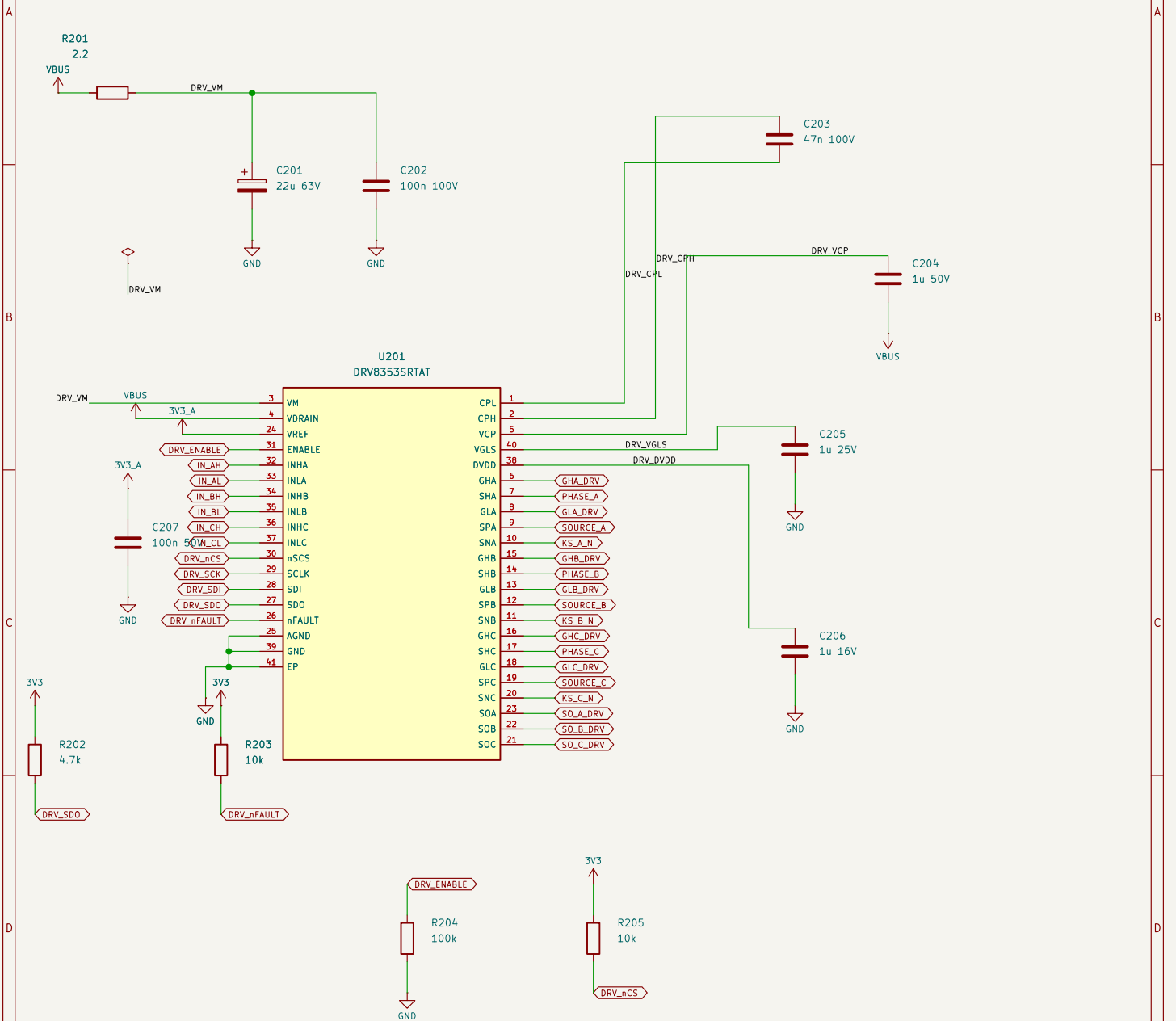
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Rev: v1-draft

Id: 12/17

Gate driver and charge pump

Gate driver / charge pump



VM filter and pump capacitors belong next to U201. VCP reservoir returns to VBUS, not GND.

SPx returns directly to each low-side source. SNx / CSA inputs use Kelvin shunt sense pads.

Six PWM inputs arrive through the hardware arm interlock. ENABLE is separate for driver setup.

Sheet: /12_gate_driver/
File: 12_gate_driver.kicad_sch

Title: Gate driver and charge pump

Size: A4
KiCad E.D.A. 10.0.4

Date:

Rev: v1-draft

Id: 13/17

USB and status indicators

USB-C device port

USB_VBUS_RAW (0.5A / 6V) → USB_5V

USB_VBUS_RAW → C413 (100n 50V) → GND

J401 USB4105-GF-A

USB_DP_PORT, USB_DM_PORT, USB_DP_ESD, USB_DM_ESD, USB_DP, USB_DM

U404 USBLC6-2SC6

R411 5.1k, R412 5.1k, R413 22, R414 22, R415 10k, R416 15k

LED_STATUS, LED_COMM, LED_3

D511 STATUS, D512 COMM, D513 POWER

TP401 5V_BUS, TP402 5V_SYS, TP403 3V3, TP404 GND

USB shield bonds to CHASSIS. D+/D- protection belongs close to the connector; 22 ohm resistors belong near the MCU.

Sheet: /13_usb_status/
File: 13_usb_status.kicad_sch
Title: USB and status indicators
Size: A3 Date: Rev: v1-draft
KiCad E.D.A. 10.0.4 Id: 14/17

USB and status indicators

USB-C device port

USB_VBUS_RAW (0.5A / 6V) → USB_5V

USB_VBUS_RAW → C413 (100n 50V) → GND

J401 USB4105-GF-A

USB_DP_PORT, USB_DM_PORT, USB_DP_ESD, USB_DM_ESD, USB_DP, USB_DM

U404 USBLC6-2SC6

R411 5.1k, R412 5.1k, R413 22, R414 22, R415 10k, R416 15k

LED_STATUS, LED_COMM, LED_3

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USB_DP_PORT, USB_DM_PORT, USB_DP_ESD, USB_DM_ESD, USB_DP, USB_DM

U404 USBLC6-2SC6

R411 5.1k, R412 5.1k, R413 22, R414 22, R415 10k, R416 15k, R511 1k, R512 1k, R513 1k

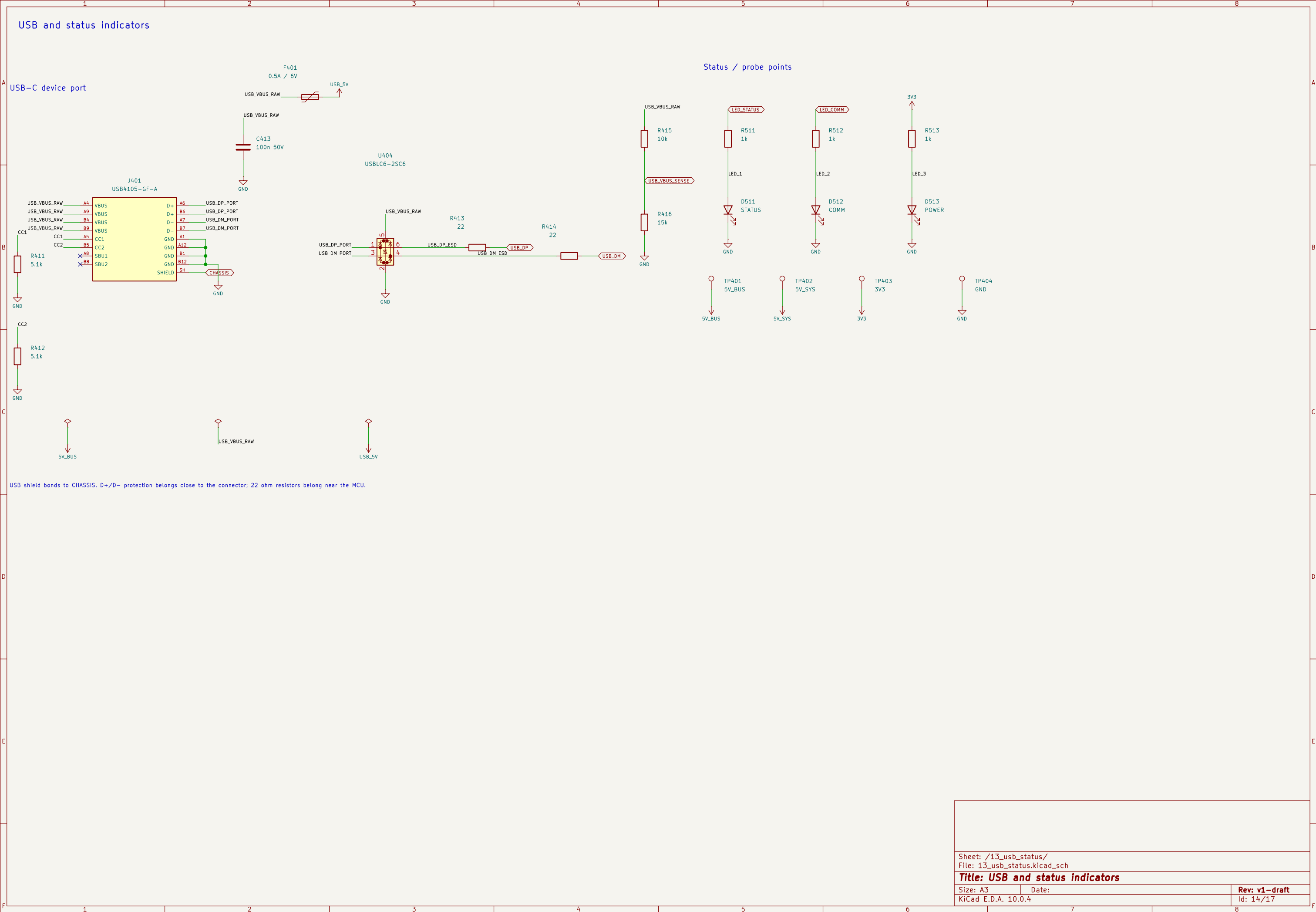
LED_STATUS, LED_COMM, LED_3

D511 STATUS, D512 COMM, D513 POWER

TP401 5V_BUS, TP402 5V_SYS, TP403 3V3, TP404 GND

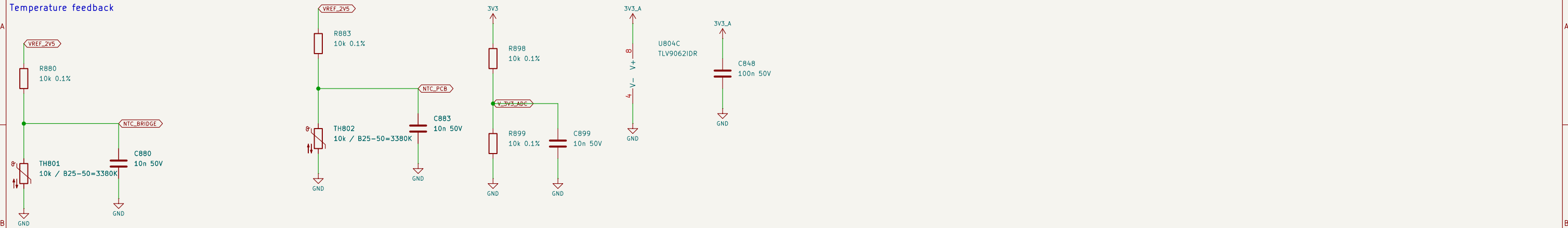
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Sheet: /13_usb_status/
File: 13_usb_status.kicad_sch
Title: USB and status indicators
Size: A3 Date: Rev: v1-draft
KiCad E.D.A. 10.0.4 Id: 14/17

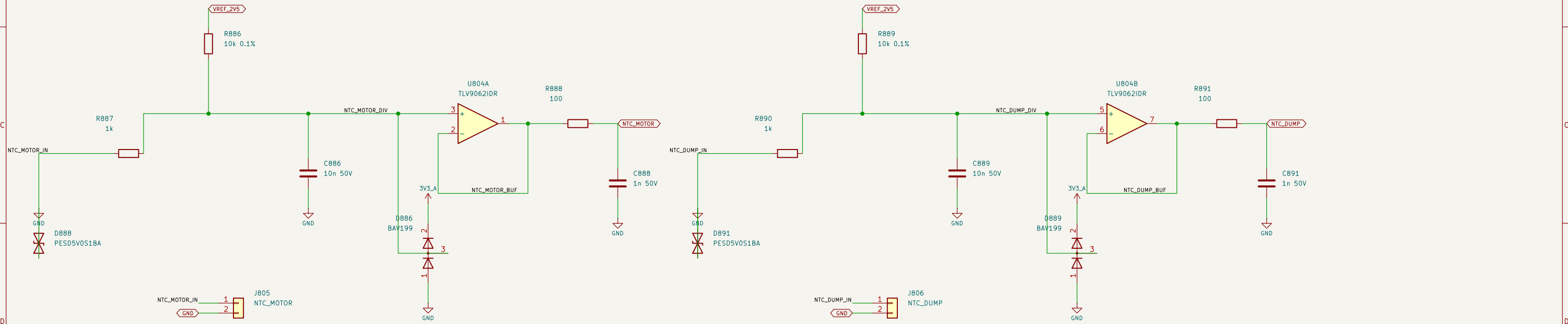
[illegible]

Temperature and logic–rail feedback

Temperature feedback

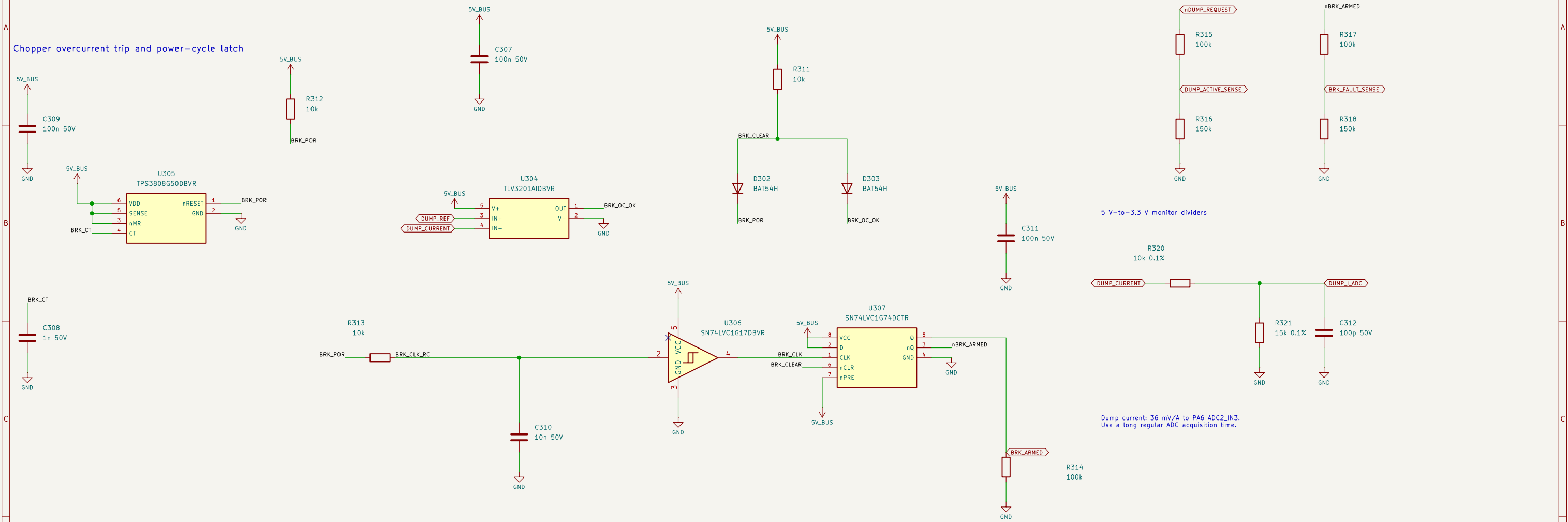


TH801/TH802: 10k, B25/50=3380 K, External NTC conversion must subtract the 1k series resistor.
External channels are buffered; configure their actual R/T curves before thermal-limit testing.



Brake protection and telemetry

Chopper overcurrent trip and power-cycle latch

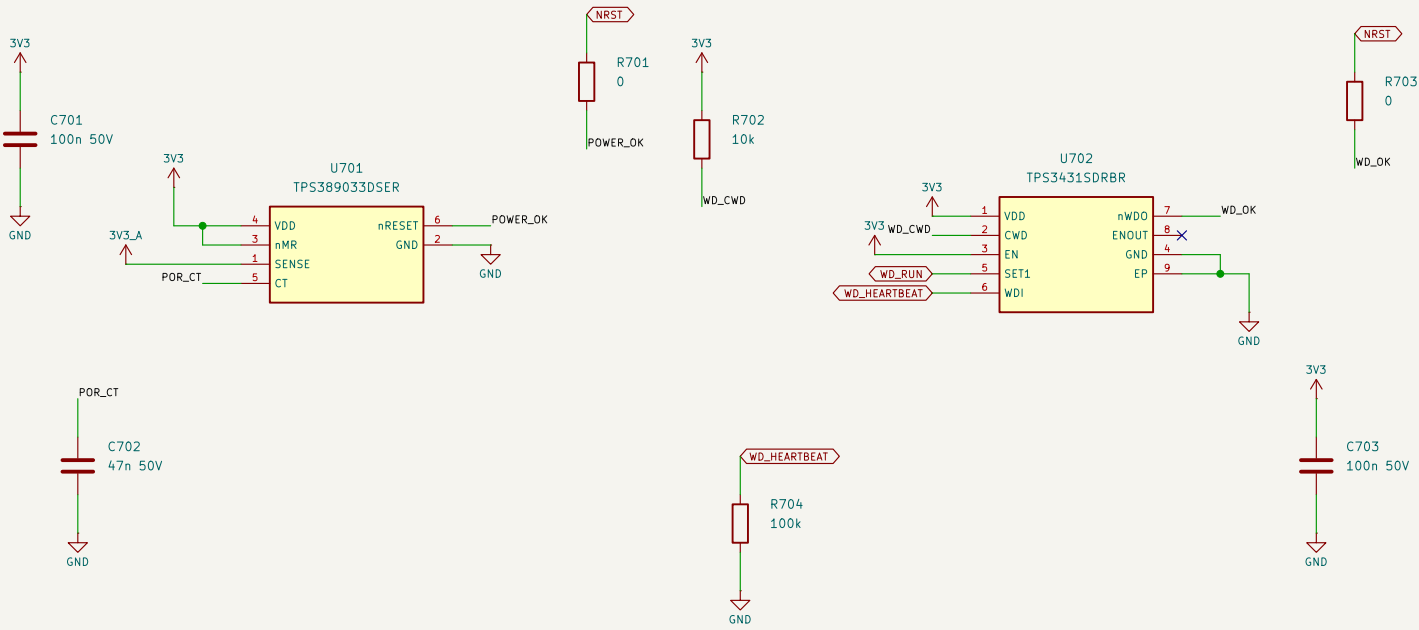


5 V-to-3.3 V monitor dividers

Dump current: 36 mV/A to PA6 ADC2_IN3.
Use a long regular ADC acquisition time.

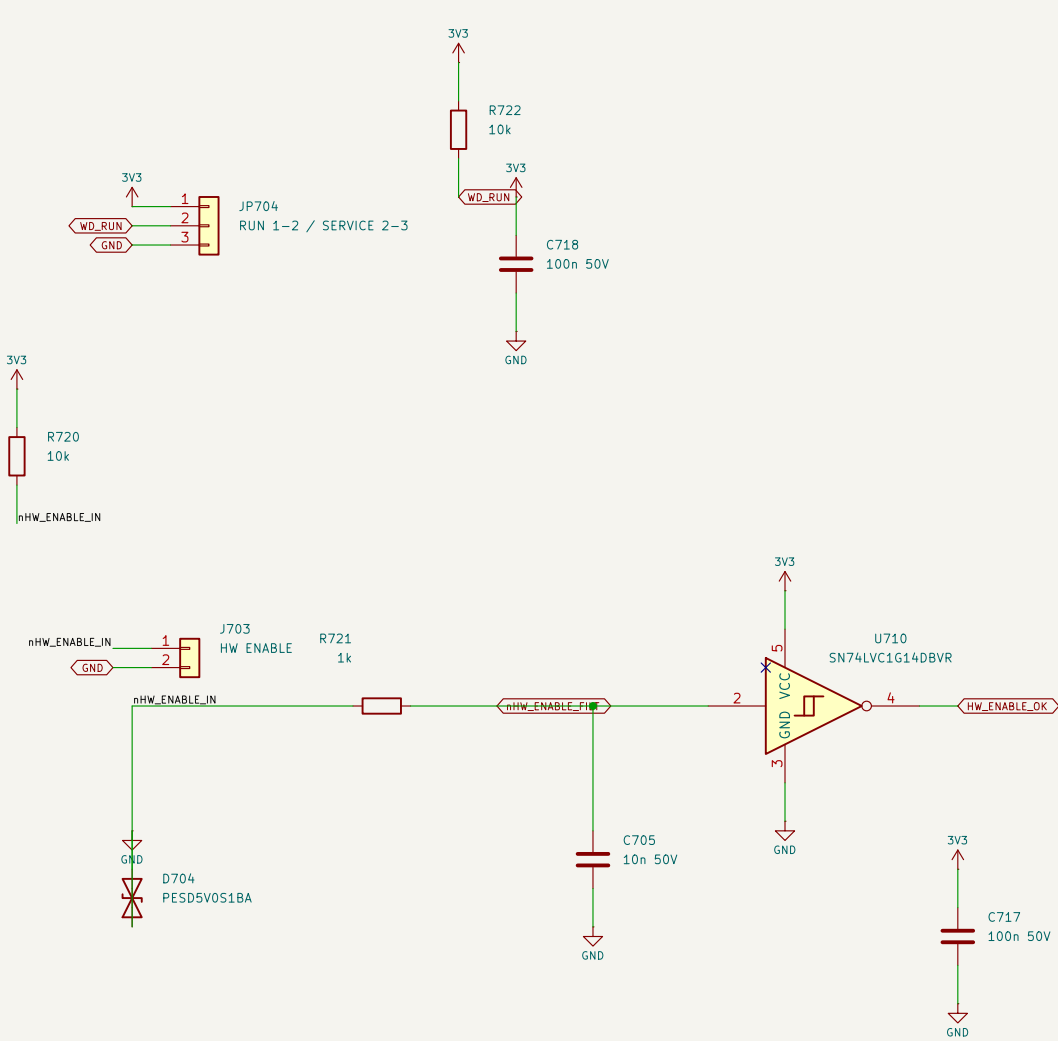
Reset, watchdog and hardware enable

Power-on reset and watchdog



Open-drain reset outputs share NRST through zero-ohm links.
R504 on the MCU sheet provides the pull-up.

Service and hardware enable



RUN 1-2 / SERVICE 2-3.
SERVICE or a chopper fault forces
both driver-enable and bridge-arm paths off.